

SECONDARY 3 EXPRESS — END-OF-YEAR EXAMINATION
ADDITIONAL MATHEMATICS 4049
WORKED SOLUTIONS & MARKING SCHEME

Question • Working • Marks & examiner's notes

Mark codes:

- M** method mark — awarded for a correct method, even if the arithmetic that follows is wrong
- A** accuracy mark — awarded for a correct answer or correct intermediate value; dependent on the preceding M mark
- B** independent mark — awarded for a correct statement, value or result on its own, with no method required

Total: 90 marks. Follow-through (ft) applies throughout: a correct method applied to a candidate's own earlier incorrect value still earns the M mark.

Question 1 — Surds — area of a triangle**[3]**

The triangle DEF has an area of $\frac{1}{4}(7\sqrt{3}+9)$ cm². The length of DE is $(\sqrt{3}+2)$ cm and angle EDF is 60° . Find, without using a calculator, the length of DF , in cm, in the form $(a+b\sqrt{3})$.

$$\text{Area} = \frac{1}{2}(DE)(DF)\sin \angle EDF, \text{ so } \frac{1}{2}(\sqrt{3}+2)(DF)\left(\frac{\sqrt{3}}{2}\right) = \frac{1}{4}(7\sqrt{3}+9) \quad \text{M1}$$

$$\sqrt{3}(\sqrt{3}+2)(DF) = 7\sqrt{3}+9 \Rightarrow (3+2\sqrt{3})(DF) = 7\sqrt{3}+9$$

$$DF = \frac{7\sqrt{3}+9}{3+2\sqrt{3}} \times \frac{3-2\sqrt{3}}{3-2\sqrt{3}} = \frac{3\sqrt{3}-15}{-3} \quad \text{M1}$$

$$\text{Answer: } DF = 5 - \sqrt{3} \text{ cm, so } a = 5 \text{ and } b = -1. \quad \text{A1}$$

Note: The $\frac{1}{2}$ and the $\sin 60^\circ = \frac{\sqrt{3}}{2}$ must both appear for the first M1. Denominator

$(3)^2 - (2\sqrt{3})^2 = 9 - 12 = -3$ is negative — candidates who drop the sign get $\sqrt{3}-5$ and lose the A1.

Question 2 — Surds — rationalising the denominator**[4]**

(a) Simplify $\frac{7}{3-\sqrt{2}}$ in the form $a+b\sqrt{2}$. [2]

$$\frac{7}{3-\sqrt{2}} \times \frac{3+\sqrt{2}}{3+\sqrt{2}} = \frac{7(3+\sqrt{2})}{9-2} = \frac{7(3+\sqrt{2})}{7} \quad \text{M1}$$

$$\text{Answer: } 3+\sqrt{2}, \text{ so } a=3, b=1. \quad \text{A1}$$

(b) Given $(2+\sqrt{3})x = 5+4\sqrt{3}$, find x in the form $a+b\sqrt{3}$. [2]

$$x = \frac{5+4\sqrt{3}}{2+\sqrt{3}} \times \frac{2-\sqrt{3}}{2-\sqrt{3}}, \text{ denominator} = 4-3=1 \quad \text{M1}$$

$$\text{Numerator} = 10 - 5\sqrt{3} + 8\sqrt{3} - 12 = -2 + 3\sqrt{3}$$

$$\text{Answer: } x = -2 + 3\sqrt{3}, \text{ so } a = -2, b = 3. \quad \text{A1}$$

Note: In (b) the denominator rationalises to exactly 1 — a deliberate check that candidates still multiply out rather than guess.

Question 3 — Compound angles**[4]**

$\sin A = \frac{-1}{\sqrt{5}}$ and $\tan B = \frac{3}{4}$, where A and B are in the same quadrant. Find p if $\cos(A - B) = p\sqrt{5}$.

$\sin A < 0 \Rightarrow A$ in quadrant 3 or 4; $\tan B > 0 \Rightarrow B$ in quadrant 1 or 3. Same quadrant $\Rightarrow$ both in quadrant 3. **B1**

Quadrant 3: $\cos A = \frac{-2}{\sqrt{5}}$ (from $\cos^2 A = 1 - \frac{1}{5}$)

$\tan B = \frac{3}{4}$ in quadrant 3 gives $\sin B = \frac{-3}{5}$, $\cos B = \frac{-4}{5}$ (3–4–5 triangle, both negative) **B1**

$\cos(A - B) = \cos A \cos B + \sin A \sin B = \left(-\frac{2}{\sqrt{5}}\right)\left(-\frac{4}{5}\right) + \left(-\frac{1}{\sqrt{5}}\right)\left(-\frac{3}{5}\right)$ **M1**

$$= \frac{8}{5\sqrt{5}} + \frac{3}{5\sqrt{5}} = \frac{11}{5\sqrt{5}} = \frac{11\sqrt{5}}{25}$$

Answer: $p = \frac{11}{25}$ **A1**

Note: Identifying quadrant 3 is the whole question — both signs negative. A candidate who takes quadrant 1 gets $p = \frac{11}{25}$ by luck of sign pairing but cannot earn the two B marks.

Question 4 — Completing the square and the discriminant**[4]**

(i) Express $y = -2x^2 - 12x + 5$ as $y = -a(x + b)^2 + c$. [2]

$$-2x^2 - 12x + 5 = -2(x^2 + 6x) + 5 = -2[(x + 3)^2 - 9] + 5$$
 M1

Answer: $y = -2(x + 3)^2 + 23$, so $a = 2$, $b = 3$, $c = 23$. **A1**

(ii) Range of k for which $-2x^2 - 12x + 5 = k$ has no two distinct roots. [2]

$$-2(x + 3)^2 + 23 = k \Rightarrow (x + 3)^2 = \frac{23 - k}{2}$$

Two distinct roots need $\frac{23 - k}{2} > 0$; so NO two distinct roots when $\frac{23 - k}{2} \leq 0$ **M1**

Answer: $k \geq 23$ **A1**

Note: Equivalent by discriminant: $2x^2 + 12x + (k - 5) = 0$, $b^2 - 4ac = 144 - 8(k - 5) \leq 0$. Accept either route. $k = 23$ **MUST** be included (equal roots are not two distinct roots) — a strict inequality loses the A1.

Question 5 — Discriminant — line meets curve**[5]**

Show that $y = x - 1$ intersects $y = nx^2 + 3x - n$ at two distinct points for all real n .

Equate: $x - 1 = nx^2 + 3x - n$ **M1**

$nx^2 + 2x + (1 - n) = 0$ **A1**

Discriminant $= 2^2 - 4(n)(1 - n) = 4 - 4n + 4n^2 = 4(n^2 - n + 1)$ **M1**

$n^2 - n + 1 = \left(n - \frac{1}{2}\right)^2 + \frac{3}{4}$ **M1**

$\left(n - \frac{1}{2}\right)^2 \geq 0$, so $n^2 - n + 1 \geq \frac{3}{4} > 0$, hence discriminant > 0 for all real n — two distinct points. Shown.

A1

Note: The final A1 needs the explicit statement that the discriminant is positive — not merely that it is non-zero. Completing the square (or showing the discriminant of $n^2 - n + 1$ is $-3 < 0$) is required; asserting positivity without justification scores M1 only.

Note: Strictly the result needs $n \neq 0$, since $n = 0$ makes the curve a straight line meeting $y = x - 1$ once. The question's wording ("the curve") presumes $n \neq 0$; do not penalise candidates either way.

Question 6 — Logarithms — change of base**[5]**

Solve $\log_2(x - 2) + 2\log_4(x - 3) = \frac{1}{2}\log_3 9$.

RHS: $\log_3 9 = 2$, so $\frac{1}{2}\log_3 9 = 1$ **B1**

$2\log_4(x - 3) = 2 \cdot \frac{\log_2(x - 3)}{\log_2 4} = 2 \cdot \frac{\log_2(x - 3)}{2} = \log_2(x - 3)$ **M1**

$\log_2(x - 2) + \log_2(x - 3) = 1 \Rightarrow \log_2[(x - 2)(x - 3)] = 1$

$(x - 2)(x - 3) = 2 \Rightarrow x^2 - 5x + 4 = 0 \Rightarrow (x - 1)(x - 4) = 0$ **M1**

$x = 1$ or $x = 4$ **A1**

Domain: need $x - 2 > 0$ AND $x - 3 > 0$, i.e. $x > 3$. Reject $x = 1$. **A1**

Answer: $x = 4$

Note: The final A1 is for the rejection — it must be justified by the domain, not simply omitted. A candidate giving both roots scores a maximum of 4 out of 5.

- (i) Explain why the form $\frac{A}{x+1} + \frac{B}{x^2+4}$ cannot be used. [2]

x^2+4 is an irreducible quadratic factor (it has no real linear factors, since $x^2 = -4$ has no real solution). **B1**

A factor of degree 2 therefore requires a numerator of degree 1, i.e. $Bx+C$, not a constant. With only A and B there are two unknowns but three coefficients (x^2 , x , constant) to match, so no such A, B exist. **B1**

- (ii) Express $\frac{3x^2+3x+10}{(x+1)(x^2+4)}$ in partial fractions. [4]

Let $\frac{3x^2+3x+10}{(x+1)(x^2+4)} = \frac{A}{x+1} + \frac{Bx+C}{x^2+4}$ **M1**

$$3x^2+3x+10 = A(x^2+4) + (Bx+C)(x+1)$$

$x = -1: 3 - 3 + 10 = A(1+4) \Rightarrow 10 = 5A \Rightarrow A = 2$ **M1**

Compare x^2 : $3 = A + B \Rightarrow B = 1$ **A1**

Compare constants: $10 = 4A + C = 8 + C \Rightarrow C = 2$ **A1**

Answer: $\frac{2}{x+1} + \frac{x+2}{x^2+4}$

Note: Check by recombining: $2(x^2+4) + (x+2)(x+1) = 2x^2+8+x^2+3x+2 = 3x^2+3x+10$. ✓

Candidates who write $\frac{B}{x^2+4}$ despite part (i) can score at most the first M1 in (ii).

$A(2,3), B(-1,-1), C(3,1)$.

(i) Show $AC \perp BC$. [3]

$$m_{AC} = \frac{1-3}{3-2} = -2 \quad \text{M1}$$

$$m_{BC} = \frac{1-(-1)}{3-(-1)} = \frac{2}{4} = \frac{1}{2} \quad \text{M1}$$

$$m_{AC} \times m_{BC} = -2 \times \frac{1}{2} = -1, \text{ therefore } AC \perp BC. \text{ Shown.} \quad \text{A1}$$

(ii) $ABCD$ is a parallelogram; find D . [2]

Diagonals of a parallelogram bisect each other, so midpoint AC = midpoint BD . M1

$$\text{Midpoint } AC = \left(\frac{2+3}{2}, \frac{3+1}{2} \right) = \left(\frac{5}{2}, 2 \right)$$

$$\frac{-1+x_D}{2} = \frac{5}{2} \Rightarrow x_D = 6; \quad \frac{-1+y_D}{2} = 2 \Rightarrow y_D = 5$$

Answer: $D(6,5)$ A1

(iii) Find the area of $ABCD$. [2]

Right angle at C , so area = $2 \times$ area of triangle $ABC = 2 \times \frac{1}{2}(AC)(BC) = (AC)(BC)$ M1

$$AC = \sqrt{1^2 + (-2)^2} = \sqrt{5}, \quad BC = \sqrt{4^2 + 2^2} = 2\sqrt{5}$$

Answer: Area = $\sqrt{5} \times 2\sqrt{5} = 10$ units² A1

Note: The shoelace formula on $A(2,3), B(-1,-1), C(3,1), D(6,5)$ gives 10 as well — accept either method. Using the perpendicularity from (i) is the intended, shorter route.

$y = a \cos 2x + b$, $a > 0$, has maximum 4 and minimum -10 for $0 \leq x \leq 2\pi$.

(i) Find a and b . [3]

Maximum: $a + b = 4$; minimum: $-a + b = -10$

M1

Adding: $2b = -6 \Rightarrow b = -3$

A1

Then $a = 4 - b = 7$

A1

(ii) State the period and the amplitude. [2]

Period $= \frac{2\pi}{2} = \pi$

B1

Amplitude $= 7$

B1

Note: Amplitude is a , \textit{not} the distance from maximum to minimum (14) and not affected by the shift b .
This is the most common error on this question.

(iii) Sketch $y = 7 \cos 2x - 3$ for $0 \leq x \leq 2\pi$. [2]

Cosine shape, 2 complete cycles over $0 \leq x \leq 2\pi$

B1

Maximum 4 at $x = 0, \pi, 2\pi$; minimum -10 at $x = \frac{\pi}{2}, \frac{3\pi}{2}$; oscillating about $y = -3$

B1

Note: Curve must start at a maximum (cosine, positive a) and the two cycles must be visibly equal in width.

Crossings of $y = -3$ occur at $x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$.

$P = 45\,000 e^{kt}$, t years after the beginning of 2015. At the beginning of 2016 the vase was valued at \$50 000.

(a) Show that $k = 0.10536$ (5 s.f.). [3]

$$t = 1, P = 50\,000: 50\,000 = 45\,000 e^k \quad \text{M1}$$

$$e^k = \frac{50\,000}{45\,000} = \frac{10}{9} \quad \text{M1}$$

$$k = \ln \frac{10}{9} = 0.1053605\dots = 0.10536 \text{ (5 s.f.)}. \text{ Shown.} \quad \text{A1}$$

(b) Find the year in which the investment triples. [4]

$$\text{Tripled: } P = 3 \times 45\,000 = 135\,000 \quad \text{M1}$$

$$135\,000 = 45\,000 e^{kt} \Rightarrow e^{kt} = 3 \quad \text{M1}$$

$$t = \frac{\ln 3}{0.1053605} = \frac{1.098612}{0.1053605} = 10.427\dots \quad \text{M1}$$

$t \approx 10.43$ years after the beginning of 2015

Answer: During 2025 A1

Note: $t = 10$ is the beginning of 2025 and $t = 11$ the beginning of 2026, so $t = 10.43$ falls inside 2025. The A1 is for the YEAR, not for t — a candidate who stops at 10.4 years scores M1 M1 M1 only.

Note: Use the exact $k = \ln \frac{10}{9}$ where possible; using the rounded 0.10536 is accepted here because the question supplied it.

(a) Term independent of x in $3x^3\left(2x - \frac{1}{4x^2}\right)^{12}$. [3]

General term of $\left(2x - \frac{1}{4x^2}\right)^{12}$: $\binom{12}{r}(2x)^{12-r}\left(-\frac{1}{4x^2}\right)^r$ M1

$$= \binom{12}{r} 2^{12-r} (-1)^r 4^{-r} x^{12-3r}$$

With the factor $3x^3$ the power of x is $3+12-3r=15-3r$; set $15-3r=0 \Rightarrow r=5$ M1

$$\text{Term} = 3 \binom{12}{5} 2^7 (-1)^5 4^{-5} = 3 \times 792 \times 128 \times \frac{-1}{1024} = -3 \times 99$$

Answer: -297 A1

Note: $\binom{12}{5} = 792$ and $\frac{128}{1024} = \frac{1}{8}$. The sign comes from $(-1)^5$ — losing it gives $+297$ and costs the A1.

(b) In $(2+mx)^{15}$ the ratio of the coefficient of x^7 to that of x^9 is $\frac{4}{7}$. Find m . [5]

Coefficient of x^7 : $\binom{15}{7} 2^8 m^7$ M1

Coefficient of x^9 : $\binom{15}{9} 2^6 m^9$ M1

$$\frac{\binom{15}{7} 2^8 m^7}{\binom{15}{9} 2^6 m^9} = \frac{4}{7}$$
 M1

$$\frac{6435}{5005} = \frac{9}{7}, \text{ so } \frac{9}{7} \cdot \frac{4}{m^2} = \frac{4}{7} \Rightarrow \frac{36}{7m^2} = \frac{4}{7}$$
 M1

$$m^2 = 9$$

Answer: $m=3$ or $m=-3$ A1

Note: Both values are required — the ratio involves m^2 , so no root can be rejected. Giving only $m=3$ scores 4 out of 5.

(a) $f(x) = x^3 + x^2 + ax + 2b$ has factor $(x+3)$; $g(x) = x^3 - 4x^2 - ax - b$ leaves remainder 42 on division by $(x-5)$. Find a and b . [5]

$$(x+3) \text{ a factor} \Rightarrow f(-3) = 0: -27 + 9 - 3a + 2b = 0$$

M1

$$2b = 3a + 18 \dots(1)$$

A1

$$g(5) = 42: 125 - 100 - 5a - b = 42$$

M1

$$b = -5a - 17 \dots(2)$$

A1

$$\text{Substituting (2) into (1): } 2(-5a - 17) = 3a + 18 \Rightarrow -13a = 52$$

$$\text{Answer: } a = -4, b = 3$$

A1

Note: Check: (1) gives $2(3) = 6$ and $3(-4) + 18 = 6$ ✓. Sign slips in $f(-3)$ ($(-3)^3 = -27$, $(-3)^2 = +9$) are the usual cause of lost A marks.

(b) Solve $x^3 - x^2 - 7x + 7 = 0$, giving the roots in exact form. [5]

$$\text{Group: } x^2(x-1) - 7(x-1) = 0$$

M1

$$(x-1)(x^2 - 7) = 0$$

A1

$$x^2 - 7 = 0 \Rightarrow x^2 = 7$$

M1

$$x = 1$$

A1

$$\text{Answer: } x = 1, x = \sqrt{7}, x = -\sqrt{7}$$

A1

Note: "Exact form" means $\pm\sqrt{7}$ must be left in surd form — giving ± 2.65 loses the final A1. Factor theorem with trial $x = 1$ is equally acceptable in place of grouping.

$L = A d^k$, with d : 9, 50, 200, 400, 900 and L : 8, 15, 25, 32, 43.

(a) Explain how a straight line graph can be drawn, and draw it. [4]

Take $\lg$ of both sides: $\lg L = \lg A + k \lg d$

B1

Comparing with $Y = mX + c$: plot $\lg L$ (vertical) against $\lg d$ (horizontal); the graph is a straight line of gradient k and vertical intercept $\lg A$.

B1

$\lg d$: 0.954, 1.699, 2.301, 2.602, 2.954

M1

$\lg L$: 0.903, 1.176, 1.398, 1.505, 1.633

A1

(b) Estimate A and k . [4]

Gradient $k = \frac{1.633 - 0.903}{2.954 - 0.954} = \frac{0.730}{2.000}$

M1

Answer: $k \approx 0.365$

A1

Intercept: $\lg A = \lg L - k \lg d = 0.903 - 0.365(0.954) = 0.555$

M1

Answer: $A = 10^{0.555} \approx 3.59$

A1

Note: These are graphical estimates: accept k in 0.35 to 0.38 and A in 3.3 to 3.9, provided the gradient is computed from two points on the DRAWN LINE and not from two raw data points.

Note: Check: $3.59 \times 200^{0.365} = 3.59 \times 6.92 \approx 24.8$, against the tabulated $L = 25$ ✓

(a) Prove $\frac{\cos \theta}{1 - \sin \theta} - \frac{1 - \sin \theta}{\cos \theta} = 2 \tan \theta$. [5]

$$\text{LHS} = \frac{\cos^2 \theta - (1 - \sin \theta)^2}{(1 - \sin \theta) \cos \theta} \quad \text{M1}$$

$$(1 - \sin \theta)^2 = 1 - 2 \sin \theta + \sin^2 \theta \quad \text{M1}$$

$$\text{Numerator} = \cos^2 \theta - 1 + 2 \sin \theta - \sin^2 \theta; \text{ using } \cos^2 \theta = 1 - \sin^2 \theta: \quad \text{M1}$$

$$= 1 - \sin^2 \theta - 1 + 2 \sin \theta - \sin^2 \theta = 2 \sin \theta - 2 \sin^2 \theta = 2 \sin \theta (1 - \sin \theta) \quad \text{M1}$$

$$\text{LHS} = \frac{2 \sin \theta (1 - \sin \theta)}{(1 - \sin \theta) \cos \theta} = \frac{2 \sin \theta}{\cos \theta} = 2 \tan \theta = \text{RHS. Proved.} \quad \text{A1}$$

Note: Work on one side only. Candidates who cross-multiply from $\text{LHS} = \text{RHS}$ have assumed the result and score a maximum of 3 (the M marks for the algebra).

(b) Hence solve $\frac{\cos \theta}{1 - \sin \theta} - \frac{1 - \sin \theta}{\cos \theta} = \cot \theta$ for $0 \leq \theta \leq 2\pi$. [4]

$$\text{By (a): } 2 \tan \theta = \cot \theta = \frac{1}{\tan \theta} \quad \text{M1}$$

$$2 \tan^2 \theta = 1 \Rightarrow \tan \theta = \pm \frac{1}{\sqrt{2}} \quad \text{M1}$$

$$\text{Basic angle } \alpha = \tan^{-1} \frac{1}{\sqrt{2}} = 0.6155 \text{ rad}$$

$$\tan \theta = \frac{+1}{\sqrt{2}}: \theta = 0.6155, \pi + 0.6155 = 3.757 \quad \text{A1}$$

$$\tan \theta = \frac{-1}{\sqrt{2}}: \theta = \pi - 0.6155 = 2.526, 2\pi - 0.6155 = 5.668$$

$$\text{Answer: } \theta = 0.615, 2.53, 3.76, 5.67 \text{ rad (3 s.f.)} \quad \text{A1}$$

Note: All four solutions are required for the final A1. The range is in radians — answers in degrees score a maximum of 3. Note $\theta \neq \frac{\pi}{2}, \frac{3\pi}{2}$ (undefined) but none of the solutions violate this.

(c) Find the set of values of k for which $\frac{\cos \theta}{1 - \sin \theta} - \frac{1 - \sin \theta}{\cos \theta} = k(1 + \tan^2 \theta)$ has no solutions. [3]

$$\text{By (a) the equation becomes } 2 \tan \theta = k(1 + \tan^2 \theta). \text{ Let } t = \tan \theta: \quad \text{M1}$$

$$k t^2 - 2t + k = 0$$

$$\text{No real } t \text{ requires discriminant } < 0: (-2)^2 - 4(k)(k) < 0 \quad \text{M1}$$

$$4 - 4k^2 < 0 \Rightarrow k^2 > 1$$

$$\text{Answer: } k < -1 \text{ or } k > 1 \text{ (equivalently } k^2 > 1) \quad \text{A1}$$

Note: $k = 0$ gives $-2t = 0$, i.e. $t = 0$, which DOES have solutions ($\theta = 0, \pi, 2\pi$) — consistent, since $0^2 < 1$.

Candidates who divide by k without considering $k = 0$ should not be penalised provided the final set is correct.